

Flash-SD-KDE TODOs

Paper

- ☐ Ensure Figures 1–2 placement is acceptable (float pages / captions).
- ☐ Tighten motivation in Introduction (one clear paragraph on the bottleneck).
- ☐ Add/verify Related Work positioning vs. prior GPU KDE / SD-KDE implementations.
- ☐ Clarify method: explicitly show the re-ordered computation and where GEMMs appear.
- ☐ Include limitations / when Tensor Cores do not help (small d , small n , non-multiple-of-16).
- ☐ Proofread for consistent terminology (SD-KDE vs. score-debiased KDE, 16-D vs. 16-dimensional).

Experiments

- ☐ Re-check headline speedups and the exact hardware/software stack.
- ☐ Add a numerical-accuracy comparison table/plot vs. baselines (relative error / ranking).
- ☐ Add scaling sweep (vary n , d , and n_{test}) and summarize trends.
- ☐ Confirm timing methodology (warmup, sync points, median over trials, confidence intervals).

Artifacts

- ☐ Ensure code release instructions reproduce all figures from a clean checkout.
- ☐ Add a minimal “quickstart” in the repo README (env, commands, expected outputs).
- ☐ Record exact commit hashes / versions for Torch, Triton, CUDA, driver.

Submission

- ☐ Confirm bibliography formatting and citation completeness.
- ☐ Run a final compile pass and check for overfull boxes / missing refs.
- ☐ Double-check author/affiliation and anonymization requirements (if any).